

Data Visualization Lab

Assignment 4

Power Query & Sales Analysis Assignment

Part A: Power Query Operations

A. Grouping

2. Find the total sales amount for each product category.

Full table was grouped by Category Name, and the Sales Amount column was summed for each group. This shows how much revenue each product category (Bikes, Accessories, etc.) brought in overall.

APPLIED STEPS			
Source			
Promoted headers			
Changed column type			
Merged Queries			
Expanded AdventureWorks Pr...			
Grouped Rows			
Merged Queries1			
Expanded AdventureWorks Pr...			
Reordered Columns			
Sorted Rows			
✕ Renamed Columns			

	1 ² ProductCategoryKey	A ^B CategoryName	1.2 Total Sales
1		1 Bikes	149514.2018
2		2 Components	57048.6776
3		3 Clothing	1773.8377
4		4 Accessories	993.4284

2. Calculate the average sales amount for each customer.

Grouped by Customer Key with Average of Sales Amount as the aggregation. This gives the typical order value for each customer

All Properties			
APPLIED STEPS			
Source			
Promoted headers			
Changed column type			
Merged Queries			
Expanded AdventureWorks S...			
Grouped Rows			
✕ Sorted Rows			

	1 ² AdventureWorks Sales Data 2022.CustomerKey	1.2 Average Sales
1	null	683.3817068
2	11001	146.90355
3	11012	3.14
4	11017	742.35
5	11018	197.49355
6	11019	24.3865875
7	11024	4.743333333
8	11025	398.17
9	11026	258.7766667
10	11027	194.655
11	11028	742.35
12	11029	252.11
13	11030	940.8046

3. Identify the top 10 customers based on total sales.

Grouped by Customer Key and summed Sales Amount, then sorted the result in descending order.

Used Keep Top Rows (10) to trim the table down to just the highest-spending customers.

APPLIED STEPS		
Source		
Promoted headers		
Changed column type		
Merged Queries		
Expanded AdventureWorks S...		
Grouped Rows		
Sorted Rows		
Kept First Rows		

1.2	AdventureWorks Sales Data 2022.CustomerKey	1.2 Total Sales
1	29480	2440.3384
2	29472	88.98
3	29470	59.1242
4	29465	791.32
5	29464	756.33
6	29463	751.34
7	29462	853.4742
8	29461	775.9942
9	29460	751.34
10	29459	804.9842

4. Find the total quantity sold for each product subcategory.

Grouped by Subcategory Name and summed Order Quantity for each group.

APPLIED STEPS		
Source		
Promoted headers		
Changed column type		
Merged Queries		
Expanded AdventureWorks S...		
Grouped Rows		
Sorted Rows		

1.2	ProductSubcategoryKey	1.2 Total Quantity Sold
1	37	16503
2	28	8698
3	31	3444
4	19	2394
5	2	2324
6	30	2222
7	1	2052
8	21	1786
9	20	1539
10	3	1313
11	29	949
12	23	580
13	22	516
14	32	394
15	25	317

5. Determine the average order quantity for each sales territory.

Grouped by Territory key and applied Average on Order Quantity.

APPLIED STEPS		
Source		
Promoted headers		
Changed column type		
Merged Queries		
Expanded AdventureWorks S...		
Merged Queries1		
Expanded AdventureWorks T...		
Grouped Rows		
Renamed Columns		

1.2	TerritoryKey	1.2 Average Order Quantity
1	1	1.535972851
2	10	1.545675266
3	4	1.53381248
4	9	1.50229612
5	6	1.595860566
6	8	1.538407576
7	7	1.53701016
8	5	1.4
9	3	1.4
10	2	1.6

6. Count the number of orders placed by each customer.

Grouped by Customer Key and used Count Rows as the aggregation to get the number of order lines per customer.

APPLIED STEPS

- Source
- Promoted headers
- Changed column type
- Merged Queries
- Expanded AdventureWorks S...
- Merged Queries1
- Expanded AdventureWorks T...
- Grouped Rows
- Sorted Rows

	1 ² AdventureWorks Sales Data 2022.CustomerKey	1 ² Number of Orders Placed
1	11185	45
2	11300	45
3	11262	40
4	11276	36
5	11223	33
6	11330	31
7	11091	30
8	11277	29
9	11711	28
10	11566	25
11	11200	23
12	11211	22
13	11331	22
14	11287	22

7. Find the highest sales amount recorded for each product category.

Grouped by Category Name and used Max on the Sales Amount column.

This shows the single biggest sale within each category.

APPLIED STEPS

- Source
- Promoted headers
- Changed column type
- Merged Queries
- Expanded AdventureWorks S...
- Merged Queries1
- Expanded AdventureWorks T...
- Added Custom
- Merged Queries2
- Expanded AdventureWorks Pr...
- Merged Queries3
- Expanded AdventureWorks Pr...
- Grouped Rows

	1 ² AdventureWorks Product Categories Lookup.ProductCategor...	A ^B _C AdventureWorks Product Categories Lookup.CategoryNa...	1.2 Sales Amount
1	4	Accessories	159
2	1	Bikes	2384.07
3	3	Clothing	70.6443
4	2	Components	null

8. Calculate the total tax amount collected for each sales category.

Grouped by Category and summed the Tax Amount column (calculated earlier as a percentage of Sales Amount). This shows how much tax revenue is associated with each Category.

Source	
Promoted headers	
Changed column type	
Merged Queries	
Expanded AdventureWorks S...	
Merged Queries1	
Expanded AdventureWorks T...	
Added Custom	
Merged Queries2	
Expanded AdventureWorks Pr...	
Merged Queries3	
Expanded AdventureWorks Pr...	
Added Custom1	
Renamed Columns	
Removed Columns	
Renamed Columns1	
Removed Columns1	
Renamed Columns2	
Removed Columns2	
Renamed Columns3	
Grouped Rows	

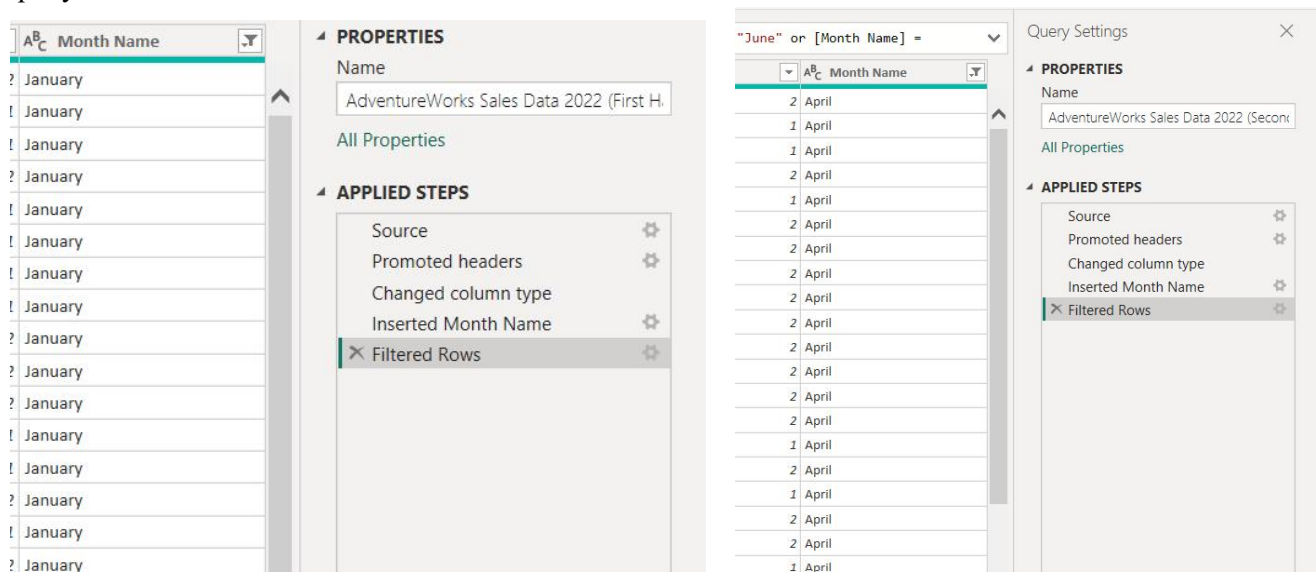
	1.2.3 ProductCategoryKey	A.8.C. CategoryName	1.2 Total Tax Amount per Sales Category
1	4	Accessories	40586.47914
2	1	Bikes	677508.3624
3	3	Clothing	16741.11425
4	2	Components	null

B. Pivoting and Unpivoting

1. Create a table containing Product Category, Sales Amount, Tax Amount, and Order Quantity.
2. Unpivot the numerical columns (Sales Amount, Tax Amount, Order Quantity) into Metric and Value columns.
3. Rename the generated columns: Attribute → Metric, Value → Amount.
4. Pivot the Metric column back to its original structure.
5. Verify that the pivoted table matches the original data.

APPLIED STEPS	
Source	
Promoted headers	
Changed column type	
Merged Queries	
Expanded AdventureWorks S...	
Merged Queries1	
Expanded AdventureWorks T...	
Added Custom	
Merged Queries2	
Expanded AdventureWorks Pr...	
Merged Queries3	
Expanded AdventureWorks Pr...	
Added Custom1	
Renamed Columns	
Removed Columns	
Renamed Columns1	
Removed Columns1	
Renamed Columns2	
Removed Columns2	
Renamed Columns3	
Reordered Columns	
Grouped Rows	
Unpivoted Columns	
Renamed Columns4	

	1.2.3 ProductCategoryKey	A.8.C. CategoryName	A.8.C. Metric	1.2 Amount
1	4	Accessories	Sales Amount	507330.9892
2	4	Accessories	Tax Amount	40586.47914
3	4	Accessories	Order Quantity	32493
4	1	Bikes	Sales Amount	8468854.53
5	1	Bikes	Tax Amount	677508.3624
6	1	Bikes	Order Quantity	5689
7	3	Clothing	Sales Amount	209263.9281
8	3	Clothing	Tax Amount	16741.11425
9	3	Clothing	Order Quantity	7132



3. Verify the total number of records after appending.

Counted rows in the First Half and Second Half queries separately, then confirmed their sum equals the row count of the appended table.

The screenshot shows the Power BI Query Editor interface. The formula bar at the top contains the expression: `List.NonNullCount(#"Changed column type"[OrderDate])`. Below the formula bar, the result area displays the number 29481. On the right side, the 'Query Settings' pane is open, showing the 'APPLIED STEPS' list. The steps listed are: Source, Promoted headers, Changed column type, and Calculated Count (which is currently selected and highlighted).

The screenshot shows the Power BI Query Editor interface. The formula bar at the top contains the expression: `List.NonNullCount(Source[OrderDate])`. Below the formula bar, the result area displays the number 29481. On the right side, the 'Query Settings' pane is open, showing the 'APPLIED STEPS' list. The steps listed are: Source and Calculated Count (which is currently selected and highlighted).

D. Merge Queries

1. Merge Sales Data with Product lookup using ProductKey.

2. Expand the merged table to include:

- Product Name
- Color
- Product Cost

3. Merge the result with ProductSubcategory.

4. Expand Product Subcategory Name.

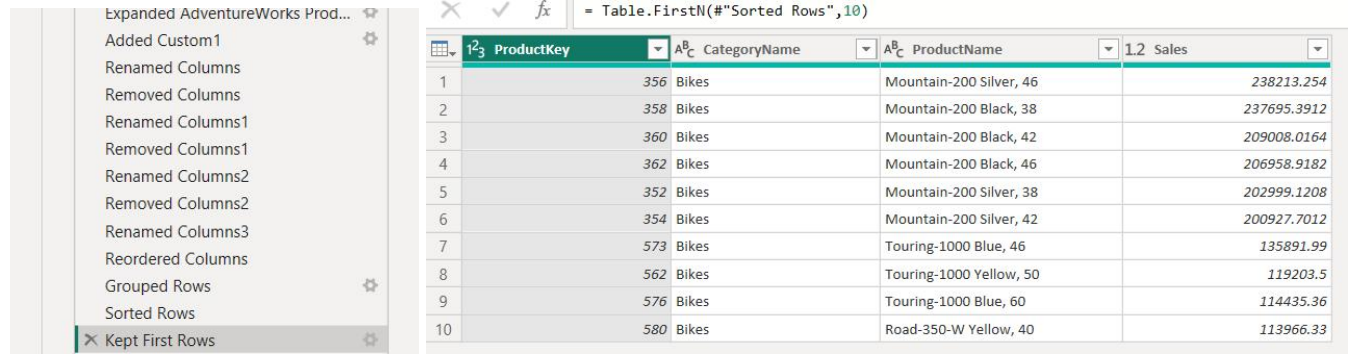
5. Merge the result with ProductCategory.

6. Expand English Product Category Name.

The screenshot shows the 'APPLIED STEPS' list in the Power BI Query Editor. The steps listed are: Source, Promoted headers, Changed column type, Merged Queries, Expanded AdventureWorks Product Lookup, Removed Columns, Merged Queries1, Expanded AdventureWorks Product Subcategories L..., Merged Queries2, and Expanded AdventureWorks Product Categories Loo... (which is currently selected and highlighted).

3. Find the Top 10 Products based on Sales Amount.

Grouped by Product Name, summed Sales Amount, sorted descending, and kept the top 10 rows.

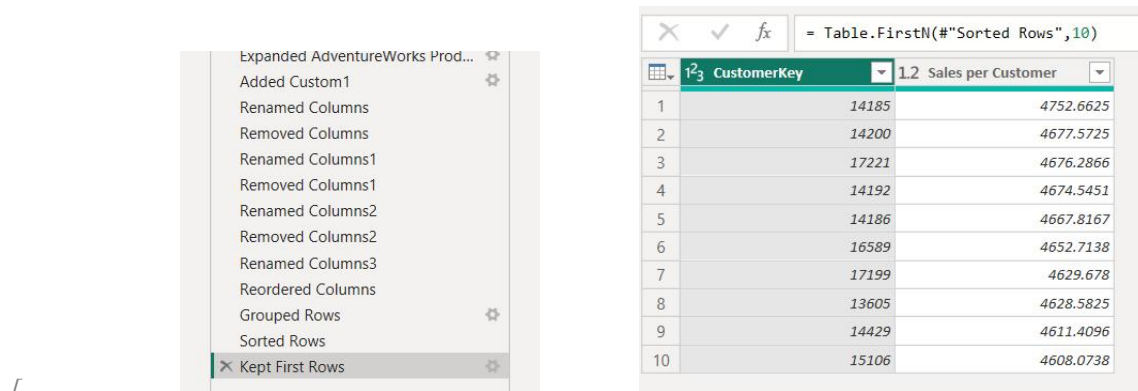


The screenshot shows the 'Kept First Rows' step selected in the Applied Steps pane. The main view displays a table with the following data:

	ProductKey	CategoryName	ProductName	Sales
1	356	Bikes	Mountain-200 Silver, 46	238213.254
2	358	Bikes	Mountain-200 Black, 38	237695.3912
3	360	Bikes	Mountain-200 Black, 42	209008.0164
4	362	Bikes	Mountain-200 Black, 46	206958.9182
5	352	Bikes	Mountain-200 Silver, 38	202999.1208
6	354	Bikes	Mountain-200 Silver, 42	200927.7012
7	573	Bikes	Touring-1000 Blue, 46	135891.99
8	562	Bikes	Touring-1000 Yellow, 50	119203.5
9	576	Bikes	Touring-1000 Blue, 60	114435.36
10	580	Bikes	Road-350-W Yellow, 40	113966.33

4. Determine the Top 10 Customers based on Sales Amount.

Grouped by Customer Key, summed Sales Amount, sorted descending, kept top 10.

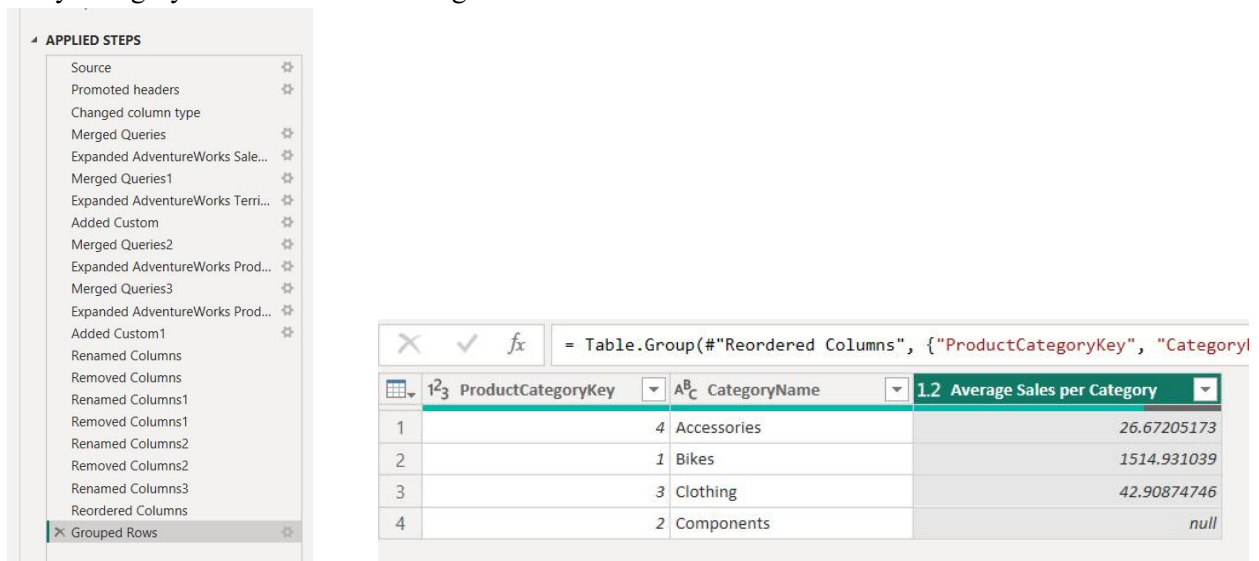


The screenshot shows the 'Kept First Rows' step selected in the Applied Steps pane. The main view displays a table with the following data:

	CustomerKey	Sales per Customer
1	14185	4752.6625
2	14200	4677.5725
3	17221	4676.2866
4	14192	4674.5451
5	14186	4667.8167
6	16589	4652.7138
7	17199	4629.678
8	13605	4628.5825
9	14429	4611.4096
10	15106	4608.0738

5. Calculate the Average Sales Amount by Product Category.

Grouped by Category Name and used Average instead of Sum on the Sales Amount column.



The screenshot shows the 'Grouped Rows' step selected in the Applied Steps pane. The main view displays a table with the following data:

	ProductCategoryKey	CategoryName	Average Sales per Category
1	4	Accessories	26.67205173
2	1	Bikes	1514.931039
3	3	Clothing	42.90874746
4	2	Components	null

6. Find the Total Order Quantity for each Product Category.

Grouped by Category Name and summed Order Quantity.

All Properties

APPLIED STEPS

Source

Promoted headers

Changed column type

Merged Queries

Expanded AdventureWorks Sale...

Merged Queries1

Expanded AdventureWorks Terri...

Added Custom

Merged Queries2

Expanded AdventureWorks Prod...

Merged Queries3

Expanded AdventureWorks Prod...

Added Custom1

Renamed Columns

Removed Columns

Renamed Columns1

Removed Columns1

Renamed Columns2

Removed Columns2

Renamed Columns3

Reordered Columns

Grouped Rows

Sorted Rows

fx

= Table.Sort("#Grouped Rows",{{"ProductCategoryKey", Order.Ascending}})

	1.2 ProductCategoryKey	A.1 CategoryName	1.2 Order Quantity
1	1	Bikes	2460
2	2	Components	null
3	3	Clothing	3365
4	4	Accessories	15350

7. Determine the highest-selling product.

Grouped by Product Name, summed Order Quantity, sorted descending, and kept the top 1 row.

Removed Columns2

Renamed Columns3

Reordered Columns

Grouped Rows

Sorted Rows

Kept First Rows

fx

= Table.FirstN("#Sorted Rows",1)

	1.2 ProductKey	A.1 ProductName	1.2 Order Quantity
1	477	Water Bottle - 30 oz.	2177

8. Identify the lowest-selling product.

Sorted the summary in ascending order by Order Quantity and took the top row.

Removed Columns2

Renamed Columns3

Reordered Columns

Grouped Rows

Sorted Rows

Filtered Rows

Kept First Rows

fx

= Table.FirstN("#Filtered Rows",1)

	1.2 ProductKey	A.1 ProductName	1.2 Order Quantity
1	586	Touring-3000 Blue, 50	7

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9. Calculate the total sales for each calendar year.

Added a custom column extracting the year from Order Date (Date.Year), then grouped by that Year column and summed Sales Amount.

Renamed Columns1	
Removed Columns1	
Renamed Columns2	
Removed Columns2	
Renamed Columns3	
Reordered Columns	
Grouped Rows	
Removed Columns3	
Merged Queries4	
Expanded Sales Data (3 years)	
Reordered Columns1	
Grouped Rows1	
Removed Blank Rows	

1.2 Year	1.2 Total Sales per Year
1 2021	80293.632
2 2022	80293.632
3 2020	28271.5534

10. Compare sales across different months.

Added a custom column extracting the month from Order Date, then grouped by month and summed Sales Amount to see seasonal patterns.

Renamed Columns3	
Reordered Columns	
Grouped Rows	
Removed Columns3	
Merged Queries4	
Expanded Sales Data (3 years)	
Reordered Columns1	
Inserted Month Name	
Reordered Columns2	
Grouped Rows1	
Removed Blank Rows	

A. Month Name	ABC Sales per Month
1 January	1163114.293
2 February	1211164.018
3 March	1288169.534
4 April	1333480.25
5 May	1509798.73
6 June	1449441.554
7 July	914173.3681
8 August	926001.4344
9 September	944079.1505
10 October	1075266.641
11 November	1038931.004
12 December	1459247.468

11. Determine which sales territory generated revenue.

Grouped Territory using Sales amount column.

4 APPLIED STEPS

- Source ✱
- Promoted headers ✱
- Changed column type
- Merged Queries ✱
- Expanded AdventureWorks Sale... ✱
- Merged Queries1 ✱
- Expanded AdventureWorks Terri... ✱
- Added Custom ✱
- Merged Queries2 ✱
- Expanded AdventureWorks Prod... ✱
- Merged Queries3 ✱
- Expanded AdventureWorks Prod... ✱
- Added Custom1 ✱
- Renamed Columns
- Removed Columns
- Renamed Columns1
- Removed Columns1
- Renamed Columns2
- Removed Columns2
- Renamed Columns3
- Reordered Columns
- Grouped Rows ✱
- ✕ Removed Blank Rows

= Table.SelectRows("#Grouped Rows", each not List.IsEmpty(List.Re

	1.2 SalesTerritoryKey	A. Country	1.2 Revenue
1	1	United States	508858.062
2	10	United Kingdom	514077.8023
3	4	United States	816547.5413
4	9	Australia	1122621.146
5	6	Canada	277211.6441
6	8	Germany	440448.7661
7	7	France	380011.4608
8	5	United States	2245.5469
9	3	United States	96.97
10	2	United States	97.1442

Grouped by Country and used Average on the Tax Amount column.

Grouped by Country and used Average on the Tax Amount column.

Advanced Tasks

Started from the Total Sales by Category table, sorted descending by Sales Amount, then added an Index Column (starting from 1) to act as the rank.

Started from the Total Sales by Category table, sorted descending by Sales Amount, then added an Index Column (starting from 1) to act as the rank.

APPLIED STEPS

- Merged Queries
- Expanded AdventureWorks S...
- Merged Queries1
- Expanded AdventureWorks T...
- Added Custom
- Merged Queries2
- Expanded AdventureWorks P...
- Merged Queries3
- Expanded AdventureWorks P...
- Added Custom1
- Renamed Columns
- Removed Columns
- Renamed Columns1
- Removed Columns1
- Renamed Columns2
- Removed Columns2
- Renamed Columns3
- Reordered Columns
- Removed Duplicates
- Removed Errors
- Grouped Rows
- Sorted Rows
- Added Index
- Reordered Columns1
- Renamed Columns4

Formula Bar: `= Table.RenameColumns("#Reordered Columns1",{{"Index", "Rank"}})`

	ProductCategoryKey	CategoryName	Total Sales	Rank
1	1	Bikes	3726730.357	1
2	4	Accessories	237567.9648	2
3	3	Clothing	97917.7617	3
4	2	Components	null	4

2. Identify products with above-average sales.

Added a custom column comparing each product's Sales Amount to an average calculated of Sales column and labelling it Above Average or Below Average.

Grouped Rows
✕ Added Conditional Column

ProductKey	Product Name	Sales	Custom
490	Short-Sleeve Classic Jersey, L	4697.13	Below/At Average
594	Mountain-500 Silver, 48	7909.86	Below/At Average
537	HL Mountain Tire	12040	Below/At Average
541	Touring Tire	14002.17	Below/At Average
489	Short-Sleeve Classic Jersey, M	4967.08	Below/At Average
371	Road-250 Red, 58	80717.8125	Above Average
356	Mountain-200 Silver, 46	238213.254	Above Average
358	Mountain-200 Black, 38	237695.3912	Above Average
360	Mountain-200 Black, 42	209008.0164	Above Average
471	Classic Vest, S	2349.5	Below/At Average
606	Road-750 Black, 52	53459.01	Above Average
566	Touring-3000 Blue, 58	9650.55	Below/At Average
385	Road-550-W Yellow, 42	52022.75	Above Average
387	Road-550-W Yellow, 44	44019.25	Above Average
560	Touring-2000 Blue, 60	25511.85	Below/At Average

3. Group customers into High, Medium, and Low spending categories using Conditional Columns.

Used the Customer total-sales summary, checked the average, and added a Conditional Column with thresholds to label each customer High, Medium, or Low.

Renamed Columns2
Removed Columns2
Renamed Columns3
Reordered Columns
Removed Duplicates
Grouped Rows
✕ Added Conditional Column

CustomerKey	Spending	Spending Categories
23791	75.57	Low
18155	53.6142	Low
12473	1277.8	High
11891	34.99	Low
11201	203.6415	Low
20683	1735.98	High
27065	131.94	Low
11464	2419.06	High
28943	64.95	Low
16717	253.1504	Low
20135	79.95	Low
11280	77.27	Low
15684	2101.3766	High
13050	1249.84	High
17184	4601.0338	High
11792	81.7115	Low
23694	103.6042	Low
20260	1734.6342	High
12563	173.6715	Low
14581	950.1804	High
16720	43.6242	Low

4. Calculate the percentage contribution of each product category to total sales.

On the Total Sales by Category table, added a custom column dividing each category's sales by total sum of the Total Sales column and multiply it by 100.

Reordered Columns
Grouped Rows
Added Custom2
✕ Renamed Columns4

= Table.RenameColumns("#Added Custom2",{{"Custom", "Percentage Contribution"}})				
123	ProductCategoryKey	ABC CategoryName	1.2 Total Sales	ABC 123 Percentage Contribution
1	4	Accessories	237567.9648	5.848235663
2	1	Bikes	3726730.357	91.74131451
3	3	Clothing	97917.7617	2.410451874
4	2	Components	null	null

5. Compare weekend vs weekday sales by extracting the day of the week from the order date.

Added a custom column using Order Date, then a Conditional Column labelling Saturday/Sunday as Weekend and the rest as Weekday.

Grouped by that Weekend/Weekday label and summed Sales Amount to compare the two.

= Table.SelectRows("#Filtered Rows", each [Day of Week] <> null and [Day of Week] <> "")				
OrderDate	Day of Week	Weekday / Weekend	Sales Amount	
01-01-2022	6	Weekend	34.99	
02-01-2022	0	Weekday	34.99	
03-01-2022	1	Weekday	34.99	
01-01-2022	6	Weekend	33.6442	
02-01-2022	0	Weekday	33.6442	
03-01-2022	1	Weekday	33.6442	
01-01-2022	6	Weekend	7.98	
02-01-2022	0	Weekday	7.98	
02-01-2022	0	Weekday	11.97	
03-01-2022	1	Weekday	11.97	
03-01-2022	1	Weekday	7.98	
01-01-2022	6	Weekend	32.6	
02-01-2022	0	Weekday	32.6	
03-01-2022	1	Weekday	32.6	
01-01-2022	6	Weekend	2181.5625	
02-01-2022	0	Weekday	2181.5625	
03-01-2022	1	Weekday	2181.5625	
01-01-2022	6	Weekend	8.6442	
01-01-2022	6	Weekend	25.9326	
01-01-2022	6	Weekend	17.2884	
02-01-2022	0	Weekday	17.2884	
02-01-2022	0	Weekday	25.9326	
03-01-2022	1	Weekday	17.2884	
03-01-2022	1	Weekday	8.6442	
02-01-2022	0	Weekday	48.0673	
01-01-2022	6	Weekend	48.0673	
03-01-2022	1	Weekday	48.0673	
01-01-2022	6	Weekend	9.98	
02-01-2022	0	Weekday	9.98	
02-01-2022	0	Weekday	14.97	

column profiling based on top 1000 rows

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= Table.Group("#Filtered Rows1", {"Weekday / Weekend"}, {"Average Sales", each List.Average([Sales Amount]), type number})	
⚠ Expression.Error: We cannot convert the value null to type Logical. Details: Value= Type=[Type]	

6. Create sales bands (Low, Medium, High) based on Sales Amount using Conditional Columns.

Add a Conditional Column with cutoff values on Sales Amount to label each row Low, Medium, or High.

Removed Columns1
Renamed Columns2
Removed Columns2
Renamed Columns3
Reordered Columns
Grouped Rows
Removed Columns3
Added Conditional Column

	123 ProductKey	ABC ProductName	ABC Sales Amount	ABC Sales Band
54	371	Road-250 Red, 58	2181.5625	High
55	356	Mountain-200 Silver, 46	2071.4196	High
56	358	Mountain-200 Black, 38	2049.0982	High
57	360	Mountain-200 Black, 42	2049.0982	High
58	471	Classic Vest, S	63.5	Low
59	606	Road-750 Black, 52	539.99	Medium
60	566	Touring-3000 Blue, 58	742.35	High
61	385	Road-550-W Yellow, 42	1000.4375	High
62	387	Road-550-W Yellow, 44	1000.4375	High
63	560	Touring-2000 Blue, 60	1214.85	High
64	539	ML Road Tire	74.97	Low
65	539	ML Road Tire	24.99	Low
66	539	ML Road Tire	49.98	Low
67	563	Touring-1000 Yellow, 54	2384.07	High
68	575	Touring-1000 Blue, 54	2384.07	High
69	561	Touring-1000 Yellow, 46	2384.07	High
70	535	LL Mountain Tire	49.98	Low
71	535	LL Mountain Tire	74.97	Low